New interactions of high-energy cosmic neutrinos: equations plan

Step 3 of the paper plan, draft of 2026-11-07, for sign-off

Nine display equations in the main text; the derivations in the appendices. The equations are taken from `notes/notes.tex` (sections named below) and rewritten in the paper's notation: M the mediator mass, g its coupling (with g' for $L_e - L_\alpha$ and g_D for the dark portal), α, β flavors, i, j mass states, a, b the eigenstates in a potential. Each check names the test or validation that verified the formula in the code.

Main text

Sec. II. New light mediators

Eq. (1): the couplings of the three families. For $L_e - L_\alpha$ ($\alpha = \mu, \tau$), the dark portal, and secret interactions through a scalar ϕ or a vector V:

$$\begin{aligned} \mathcal{L}_{L_e - L_\alpha} &= -g' Z'_\mu \left(\bar{e}\gamma^\mu e - \bar{\ell}_\alpha \gamma^\mu \ell_\alpha + \bar{\nu}_e \gamma^\mu P_L \nu_e - \bar{\nu}_\alpha \gamma^\mu P_L \nu_\alpha \right) , \\ \mathcal{L}_{\mathrm{DP}} &= -Z_{D\mu} \sum_{\alpha\beta} G_{\alpha\beta}\, \bar{\nu}_\alpha \gamma^\mu P_L \nu_\beta , \qquad G_{\alpha\beta} = g_D\, U_{\alpha 4} U^*_{\beta 4} , \\ \mathcal{L}_{\mathrm{SI}} &= -\tfrac{1}{2} \sum_{\alpha\beta} G_{\alpha\beta}\, \bar{\nu}_\alpha \nu_\beta\, \phi \quad \text{or} \quad - \sum_{\alpha\beta} G_{\alpha\beta}\, \bar{\nu}_\alpha \gamma^\mu P_L \nu_\beta\, V_\mu . \end{aligned} \tag{1}$$

In-line after it: the dark portal's mediator reaches the active neutrinos only through their mixing $U_{\alpha 4}$ with the dark neutrino; its Z mixing is set to zero (the potential planes were dropped, 2026-10-24); the secret-interaction textures: G proportional to $\mathrm{diag}(1,0,0)$, $(0,1,0)$, $(0,0,1)$ or the identity; Majorana or Dirac neutrinos.

Check: definitions; the Majorana mass-basis couplings $\lambda = U^T G U$ (notes, "Model and cross sections"; tests).

Open: the Dirac forms (notes, "Dirac neutrinos and a vector mediator"); whether to keep the dark portal's Z-mixing term, with $\kappa = \epsilon' g / c_W$ added to every flavor, as a remark (check the factor against `dp_absorption.dark_portal_coupling`).

Eq. (2): the potential of the electrons, for $L_e - L_\alpha$.

$$\mathbf{V}(\mathbf{r}) = g'^2\, \Phi_e(\mathbf{r})\, \mathrm{diag}(1, -1, 0) , \qquad \Phi_e(\mathbf{r}) = \int d^3 r'\, n_e(\mathbf{r}')\, \frac{e^{-M|\mathbf{r}-\mathbf{r}'|}}{4\pi |\mathbf{r}-\mathbf{r}'|} , \tag{2}$$

with $\mathrm{diag}(1, 0, -1)$ for $L_e - L_\tau$, and $-\mathbf{V}^*$ for antineutrinos.

In-line after it: the bodies that source it: the Universe's electrons (uniform, out to the Hubble radius), the Milky Way's (stars, cold and hot gas), the Sun's, the Moon's and the Earth's; each switches on when $1/M$ exceeds its distance.

Check: the Earth's Φ_e as its exact Yukawa integral (notes, "The Earth's own potential"); potential-dominated closed form to 10^{-11}.

Open: the overall normalization against the code's `Coef`, on one test point.

Eq. (3): where the models hold. The mediator's longitudinal mode couples like a scalar with Yukawa couplings

$$Y_{ij} = \frac{i}{M} \left(G'_{ij}\, m_i + G'_{ji}\, m_j \right) , \tag{3}$$

with G' the couplings in the mass basis; tree-level perturbative unitarity requires the largest partial wave $|\mathrm{Re}\, a_J| \leq 1/2$.

In-line after it: for $L_e - L_\mu$ this excludes $g' > 10^{1.76}\, M/\mathrm{eV}$ (lightest mass 10^{-6} eV); hatched in the figures.

Check: `tests/test_unitarity.py` reproduces Allwicher *et al.*'s Eqs. (4.7), (6.2), (6.4).

Sec. III. The trip

Eq. (4): the states that are followed.

$$H(E,z) = \frac{U\,\mathrm{diag}(m_1^2, m_2^2, m_3^2)\,U^\dagger}{2E(1+z)} + \mathbf{V}(z)\ , \tag{4}$$

for neutrinos ($U \to U^*$, $\mathbf{V} \to -\mathbf{V}^*$ for antineutrinos); its eigenstates $|a\rangle$, with flavor content $|\langle \nu_\alpha | a\rangle|^2$.

In-line after it: the potential changes over a Hubble time, against oscillation lengths of at most parsecs, so each state keeps its population; only collisions and decays move neutrinos between states.

Check: closed form where the potential dominates, to 10^{-11}.

Eq. (5): the cosmological transport. For the population $F_a(E,z)$ of state a, at observed energy E, with $dt = -dz/[(1+z)H(z)]$:

$$\frac{\partial F_a}{\partial t} = \mathcal{S}_a - \Gamma_a^{\mathrm{abs}} F_a + \mathcal{R}_a + \sum_{b\neq a} (R_{ab}F_b - R_{ba}F_a) + \frac{\partial}{\partial E}\,(b_a F_a) - \Gamma_a^{\mathrm{dec}} F_a + \mathcal{D}_a\ , \tag{5}$$

sources, absorption on the relics and its secondaries $\mathcal{R}_a$, soft scatterings, the energy lost to relics that change mass, decays and their daughters $\mathcal{D}_a$.

In-line after it: the same for antineutrinos; the two kinds are coupled through the secondaries.

Check: number conservation without absorption and decay; no effect for $U_{\alpha 4} \to 0$; the Z-burst normalization (notes, "The normalization of the resonant absorption").

Open: whether the secondaries' spectra appear here or only in the appendix.

Eq. (6): the absorption on the relics.

$$\Gamma_a^{\mathrm{abs}}(E,z) = \sum_r \Gamma_r(E,z)\,\frac{1}{2}\left|(U^\dagger G W)_{ra}\right|^2\ , \tag{6}$$

with W the eigenvectors of Eq. (4) and Γ_r the Breit–Wigner rate on relic mass state r, averaged over its thermal momenta.

In-line after it: the resonance energy, $E_{\mathrm{res}} = M^2/(2m_r)$: 1.0 and 5.76 PeV for $M = 10$ MeV, on ν_3 and ν_2.

Check: thermal kernel against brute force ($\leq 1.8 \times 10^{-4}$); the Dirac relics' helicity against brute force.

Eq. (7): from the Sun to the detector.

$$\frac{d\rho}{dx} = -i\,[H(x), \rho]\ , \qquad \rho = \sum_a F_a\,|a\rangle\langle a| \text{ at the Sun}\ , \tag{7}$$

with $H(x)$ that of Eq. (4) at $z = 0$ plus the Standard Model's matter potential.

In-line after it: the flux of flavor α at the detector is $\langle \nu_\alpha | \rho | \nu_\alpha \rangle$, averaged over the Earth's chords.

Check: unitarity of the propagator; the Sun's $L_e - L_\mu$ potential maps ν_2 onto ν_τ (in the tests of the flux at each stop of the trip).

Sec. IV. Data and statistical method

Eq. (8): the expected events. The number of events in bin k of a data set, from the flux Φ_α of each flavor and kind at the detector,

$$\mu_k = T \sum_\alpha \int dE\, d\Omega\; \Phi_\alpha(E,\Omega)\, A_\alpha^{(k)}(E,\Omega) + \mu_k^{\rm bkg}\ , \tag{8}$$

with $A_\alpha^{(k)}$ the effective area into bin k and T the exposure.

Check: HESE expectation exact (2026-10-14); DR2 tracks against the events (`dr2_validation.py`).

Open: the precise forms (HESE topologies; DR2 smearing) from the notes, "Track rates with IceTracks-DR2" and "Starting events with HESE".

Eq. (9): how a limit is read.

$$\Delta\chi^2(M,g) = \chi^2(M,g) - \min_g \chi^2(M,g)\ , \qquad \chi^2 = \min_{\boldsymbol{\eta}} \left[-2\ln \mathcal{L}(M,g,\boldsymbol{\eta})\right]\ , \tag{9}$$

with $\boldsymbol{\eta}$ the nuisance parameters (source spectrum and composition, atmospheric fluxes, lightest neutrino mass); a coupling is disfavored at 95% C.L. when $\Delta\chi^2 \geq 3.84$.

Check: coverage tests: planned (statistical-soundness plan).

In-line, Sec. V (for Fig. 1)

The earlier standard: the flavor composition at Earth from the potential at the detector alone, $f_{\beta,\oplus} = \sum_\alpha f_{\alpha,S} \sum_a |W_{\alpha a}|^2 |W_{\beta a}|^2$, with W the eigenvectors at the total potential at the detector.

Check: `tests/test_trips.py`: it returns the source composition where that potential dominates.

Appendices

- The transport equation in full: every secondary spectrum; the Galactic transport and emission; the source term (notes, "The transport equation in full").
- The thermal absorption rates and the Dirac relics' helicity.
- Decays in a medium; where the transport leaves them out.
- The secret-interaction transport, all channels (notes, "Transport").
- The Earth's chords and the zenith average.